Synthetic Assay Validation Review

A risk-oriented validation matrix for a fictional assay packet.

Validation is modeled as a set of separable checks. The matrix avoids a single “validated” label because a precise result can still be poorly controlled, and a linear readout can still be unsuitable for a given operational purpose.

Validation matrix

Dimension	Synthetic evidence	Acceptance convention	Residual risk
Precision	paired wells; replicate spread	CV <=10%	local handling variance
Recovery	fictional spiked controls	90-110%	matrix transfer mismatch
Blank response	blank wells before subtraction	stable low baseline	carryover or contamination
Linearity	staged standards	fit reviewed separately	range extrapolation
Traceability	named run and exception note	complete handoff	unexplained edit

How to read the matrix

Each row contributes evidence for a different operational risk. A pass in one row cannot repair an absent record in another. The matrix is a fictional planning aid and contains no clinical sample data or external citations.

Acceptance and escalation model

A compact review model rather than a universal rule set.

Observed pattern	Interpretation boundary	Synthetic escalation
Precision acceptable; blank unstable	Repeatability alone is insufficient	inspect blank preparation and rerun controls
Recovery outside range; linearity stable	Instrument fit does not explain recovery	review dilution and control transfer
Blank acceptable; traceability missing	Clean signal is not a complete record	hold review until lineage is restored
All checks documented	Operational readiness only	approve internal synthetic comparison

Conclusion

The value of the synthetic matrix is its refusal to collapse evidence. A reviewer can name the failing dimension, retain the associated record, and select a proportionate next step without turning a planning packet into a claim about real-world assay validity.